TABLE  CREATION CODE  :

```
-- ===================== USERS =====================
CREATE TABLE USERS (
user_id VARCHAR2(10) PRIMARY KEY,
name VARCHAR2(100) NOT NULL,
contact VARCHAR2(20) UNIQUE NOT NULL,
password VARCHAR2(100) NOT NULL,
role VARCHAR2(10) NOT NULL
CHECK (role IN ('STUDENT', 'TEACHER'))
);

-- ===================== STUDENTS =====================
CREATE TABLE STUDENTS (
student_id VARCHAR2(10) PRIMARY KEY,
user_id VARCHAR2(10) UNIQUE NOT NULL,

FOREIGN KEY (user_id)
REFERENCES USERS(user_id)
);

-- ===================== TEACHERS =====================
CREATE TABLE TEACHERS (
teacher_id VARCHAR2(10) PRIMARY KEY,
user_id VARCHAR2(10) UNIQUE NOT NULL,

FOREIGN KEY (user_id)
REFERENCES USERS(user_id)
);

-- ===================== SUBJECTS =====================
CREATE TABLE SUBJECTS (
subject_id VARCHAR2(10) PRIMARY KEY,
subject_name VARCHAR2(100) UNIQUE NOT NULL,
teacher_id VARCHAR2(10) NOT NULL,

FOREIGN KEY (teacher_id)
REFERENCES TEACHERS(teacher_id)
);

-- ===================== QUESTIONS =====================
CREATE TABLE QUESTIONS (
question_id VARCHAR2(10) PRIMARY KEY,
subject_id VARCHAR2(10) NOT NULL,
question_text VARCHAR2(500) NOT NULL,
```

```
import_source VARCHAR2(100) DEFAULT 'MANUAL',
correct_option_id VARCHAR2(10),

FOREIGN KEY (subject_id)
REFERENCES SUBJECTS(subject_id)
);

-- ===================== OPTIONS =====================
CREATE TABLE OPTIONS (
option_id VARCHAR2(10) PRIMARY KEY,
question_id VARCHAR2(10) NOT NULL,
option_label CHAR(1) NOT NULL
CHECK (option_label IN ('A','B','C','D')),
option_text VARCHAR2(200) NOT NULL,

FOREIGN KEY (question_id)
REFERENCES QUESTIONS(question_id),

UNIQUE (question_id, option_label),
UNIQUE (question_id, option_id)
);

-- ===== Correct option must belong to same question =====
ALTER TABLE QUESTIONS
ADD CONSTRAINT fk_correct_option
FOREIGN KEY (question_id, correct_option_id)
REFERENCES OPTIONS(question_id, option_id);

-- ===================== QUIZZES =====================
CREATE TABLE QUIZZES (
quiz_id VARCHAR2(10) PRIMARY KEY,
student_id VARCHAR2(10) NOT NULL,
subject_id VARCHAR2(10) NOT NULL,
quiz_date DATE DEFAULT SYSDATE,
score NUMBER(5) NOT NULL,
total_marks NUMBER(5) NOT NULL,

CHECK (score <= total_marks),

FOREIGN KEY (student_id)
REFERENCES STUDENTS(student_id),

FOREIGN KEY (subject_id)
REFERENCES SUBJECTS(subject_id)
```

```
);

-- ===================== QUIZ_ANSWERS =====================
CREATE TABLE QUIZ_ANSWERS (
answer_id VARCHAR2(10) PRIMARY KEY,
quiz_id VARCHAR2(10) NOT NULL,
question_id VARCHAR2(10) NOT NULL,
selected_option_id VARCHAR2(10),
is_correct CHAR(1) NOT NULL
CHECK (is_correct IN ('Y','N')),

FOREIGN KEY (quiz_id)
REFERENCES QUIZZES(quiz_id),

FOREIGN KEY (question_id)
REFERENCES QUESTIONS(question_id),

FOREIGN KEY (selected_option_id)
REFERENCES OPTIONS(option_id),

UNIQUE (quiz_id, question_id)
);

-- ===================== RESULTS =====================
CREATE TABLE RESULTS (
result_id VARCHAR2(10) PRIMARY KEY,
student_id VARCHAR2(10) NOT NULL,
subject_id VARCHAR2(10) NOT NULL,
quiz_id VARCHAR2(10) NOT NULL,
score NUMBER(5) NOT NULL,
total_marks NUMBER(5) NOT NULL,
result_date DATE DEFAULT SYSDATE,

FOREIGN KEY (student_id)
REFERENCES STUDENTS(student_id),

FOREIGN KEY (subject_id)
REFERENCES SUBJECTS(subject_id),

FOREIGN KEY (quiz_id)
REFERENCES QUIZZES(quiz_id)
);

-- ===================== CGPA_RECORDS =====================
```

```
CREATE TABLE CGPA_RECORDS (
cgpa_id VARCHAR2(10) PRIMARY KEY,
student_id VARCHAR2(10) NOT NULL,
calc_mode VARCHAR2(20) NOT NULL
CHECK (calc_mode IN ('MARKS_BASED','SUBJECT_BASED')),
calculated_gpa NUMBER(4,2) NOT NULL,
calc_date DATE DEFAULT SYSDATE,

FOREIGN KEY (student_id)
REFERENCES STUDENTS(student_id)
);

-- ===================== CGPA_SUBJECTS =====================
CREATE TABLE CGPA_SUBJECTS (
cgpa_sub_id VARCHAR2(10) PRIMARY KEY,
cgpa_id VARCHAR2(10) NOT NULL,
subject_id VARCHAR2(10),
marks NUMBER(5,2),
credit_hours NUMBER(3,1),
grade_point NUMBER(3,2) NOT NULL,

FOREIGN KEY (cgpa_id)
REFERENCES CGPA_RECORDS(cgpa_id),

FOREIGN KEY (subject_id)
REFERENCES SUBJECTS(subject_id)
);

-- ===================== INDEXES =====================
CREATE INDEX idx_quiz_student
ON QUIZZES(student_id);

CREATE INDEX idx_question_subject
ON QUESTIONS(subject_id);
```

INSERT:

```
INSERT INTO USERS VALUES ('U001','Sifatur Rahman Imran','01711000001','S001','STUDENT');
INSERT INTO USERS VALUES ('U002','Ratul Hasan Rafi','01711000002','S002','STUDENT');
INSERT INTO USERS VALUES ('U003','Md Hasan Al Tarek Palash','01711000003','S003','STUDENT');
INSERT INTO USERS VALUES ('U004','Md Abdullah AL Noman','01711000004','S004','STUDENT');
INSERT INTO USERS VALUES ('U005','Abu Suffian Hemel','01711000005','S005','STUDENT');
INSERT INTO USERS VALUES ('U006','Fatema Begum','01711000006','S006','STUDENT');
INSERT INTO USERS VALUES ('U007','Nusrat Jahan','01711000007','S007','STUDENT');
INSERT INTO USERS VALUES ('U008','Tanvir Ahmed','01711000008','S008','STUDENT');
INSERT INTO USERS VALUES ('U009','Mehedi Hasan','01711000009','S009','STUDENT');
INSERT INTO USERS VALUES ('U010','Sabbir Rahman','01711000010','S010','STUDENT');
INSERT INTO USERS VALUES ('U011','Tasnia Akter','01711000011','S011','STUDENT');
INSERT INTO USERS VALUES ('U012','Rifat Islam','01711000012','S012','STUDENT');
INSERT INTO USERS VALUES ('U013','Sumaiya Khatun','01711000013','S013','STUDENT');
INSERT INTO USERS VALUES ('U014','Arif Hossain','01711000014','S014','STUDENT');
INSERT INTO USERS VALUES ('U015','Mithila Roy','01711000015','S015','STUDENT');
INSERT INTO USERS VALUES ('U016','Dr. Karim Uddin','01811000001','teach01','TEACHER');
INSERT INTO USERS VALUES ('U017','Prof. Sultana Akter','01811000002','teach02','TEACHER');
INSERT INTO USERS VALUES ('U018','Mr. Jahangir Alam','01811000003','teach03','TEACHER');
INSERT INTO USERS VALUES ('U019','Ms. Roksana Parvin','01811000004','teach04','TEACHER');
INSERT INTO USERS VALUES ('U020','Dr. Imtiaz Hossain','01811000005','teach05','TEACHER');
COMMIT;
```

```
INSERT INTO STUDENTS VALUES ('S001','U001');
INSERT INTO STUDENTS VALUES ('S002','U002');
INSERT INTO STUDENTS VALUES ('S003','U003');
INSERT INTO STUDENTS VALUES ('S004','U004');
INSERT INTO STUDENTS VALUES ('S005','U005');
INSERT INTO STUDENTS VALUES ('S006','U006');
INSERT INTO STUDENTS VALUES ('S007','U007');
INSERT INTO STUDENTS VALUES ('S008','U008');
INSERT INTO STUDENTS VALUES ('S009','U009');
INSERT INTO STUDENTS VALUES ('S010','U010');
INSERT INTO STUDENTS VALUES ('S011','U011');
INSERT INTO STUDENTS VALUES ('S012','U012');
INSERT INTO STUDENTS VALUES ('S013','U013');
INSERT INTO STUDENTS VALUES ('S014','U014');
INSERT INTO STUDENTS VALUES ('S015','U015');
COMMIT;
```

```
INSERT INTO TEACHERS VALUES ('T001','U016');
INSERT INTO TEACHERS VALUES ('T002','U017');
INSERT INTO TEACHERS VALUES ('T003','U018');
INSERT INTO TEACHERS VALUES ('T004','U019');
INSERT INTO TEACHERS VALUES ('T005','U020');
COMMIT;
```

```
INSERT INTO SUBJECTS VALUES ('SUB001','Mathematics','T001');
INSERT INTO SUBJECTS VALUES ('SUB002','Physics','T002');
INSERT INTO SUBJECTS VALUES ('SUB003','Chemistry','T003');
INSERT INTO SUBJECTS VALUES ('SUB004','English','T004');
INSERT INTO SUBJECTS VALUES ('SUB005','Computer Science','T005');
INSERT INTO SUBJECTS VALUES ('SUB006','Biology','T002');
INSERT INTO SUBJECTS VALUES ('SUB007','Statistics','T001');
INSERT INTO SUBJECTS VALUES ('SUB008','Database Systems','T005');
COMMIT;
```

```
INSERT INTO QUESTIONS VALUES ('Q001','SUB001','What is the value of pi (approx)?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q002','SUB001','What is 12 squared?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q003','SUB001','What is the derivative of x^2?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q004','SUB001','What is the integral of 1/x?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q005','SUB001','What is 7 factorial (7!)?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q006','SUB002','What is the unit of force?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q007','SUB002','Speed of light in vacuum (m/s)?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q008','SUB002','What is Newton first law about?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q009','SUB002','What is the unit of electric
current?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q010','SUB002','What is Ohms law formula?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q011','SUB003','What is the atomic number of
Carbon?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q012','SUB003','What is the chemical formula of
water?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q013','SUB003','What is the pH of pure water?','MANUAL',NULL);
INSERT INTO QUESTIONS VALUES ('Q014','SUB005','What does CPU stand for?','MANUAL',NULL);
```

```
INSERT INTO QUESTIONS VALUES ('Q015','SUB005','Which language is known as mother of all languages?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q016','SUB005','What is the binary of decimal 10?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q017','SUB005','What does OOP stand for?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q018','SUB005','What is a primary key?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q019','SUB008','What does SQL stand for?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q020','SUB008','Which SQL clause filters grouped rows?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q021','SUB008','What is a foreign key?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q022','SUB008','What is normalization?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q023','SUB008','What is a JOIN in SQL?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q024','SUB008','Which normal form removes transitive dependency?','MANUAL',NULL);

INSERT INTO QUESTIONS VALUES ('Q025','SUB008','What does ACID stand for in databases?','MANUAL',NULL);

COMMIT;



INSERT INTO OPTIONS VALUES ('O001','Q001','A','3.14159');

INSERT INTO OPTIONS VALUES ('O002','Q001','B','2.71828');

INSERT INTO OPTIONS VALUES ('O003','Q001','C','1.41421');

INSERT INTO OPTIONS VALUES ('O004','Q001','D','1.73205');


INSERT INTO OPTIONS VALUES ('O005','Q002','A','124');

INSERT INTO OPTIONS VALUES ('O006','Q002','B','144');

INSERT INTO OPTIONS VALUES ('O007','Q002','C','164');

INSERT INTO OPTIONS VALUES ('O008','Q002','D','184');


INSERT INTO OPTIONS VALUES ('O009','Q003','A','x');
```

```
INSERT INTO OPTIONS VALUES ('O010','Q003','B','2x');
INSERT INTO OPTIONS VALUES ('O011','Q003','C','x^2');
INSERT INTO OPTIONS VALUES ('O012','Q003','D','2x^2');

INSERT INTO OPTIONS VALUES ('O013','Q004','A','x^2/2');
INSERT INTO OPTIONS VALUES ('O014','Q004','B','ln(x)');
INSERT INTO OPTIONS VALUES ('O015','Q004','C','1/x^2');
INSERT INTO OPTIONS VALUES ('O016','Q004','D','e^x');

INSERT INTO OPTIONS VALUES ('O017','Q005','A','720');
INSERT INTO OPTIONS VALUES ('O018','Q005','B','5040');
INSERT INTO OPTIONS VALUES ('O019','Q005','C','40320');
INSERT INTO OPTIONS VALUES ('O020','Q005','D','120');

INSERT INTO OPTIONS VALUES ('O021','Q006','A','Joule');
INSERT INTO OPTIONS VALUES ('O022','Q006','B','Newton');
INSERT INTO OPTIONS VALUES ('O023','Q006','C','Pascal');
INSERT INTO OPTIONS VALUES ('O024','Q006','D','Watt');

INSERT INTO OPTIONS VALUES ('O025','Q007','A','3x10^6');
INSERT INTO OPTIONS VALUES ('O026','Q007','B','3x10^8');
INSERT INTO OPTIONS VALUES ('O027','Q007','C','3x10^10');
INSERT INTO OPTIONS VALUES ('O028','Q007','D','3x10^5');

INSERT INTO OPTIONS VALUES ('O029','Q008','A','Gravity');
INSERT INTO OPTIONS VALUES ('O030','Q008','B','Inertia');
INSERT INTO OPTIONS VALUES ('O031','Q008','C','Momentum');
INSERT INTO OPTIONS VALUES ('O032','Q008','D','Acceleration');
```

```
INSERT INTO OPTIONS VALUES ('O033','Q009','A','Volt');
INSERT INTO OPTIONS VALUES ('O034','Q009','B','Ampere');
INSERT INTO OPTIONS VALUES ('O035','Q009','C','Ohm');
INSERT INTO OPTIONS VALUES ('O036','Q009','D','Watt');

INSERT INTO OPTIONS VALUES ('O037','Q010','A','P=IV');
INSERT INTO OPTIONS VALUES ('O038','Q010','B','V=IR');
INSERT INTO OPTIONS VALUES ('O039','Q010','C','F=ma');
INSERT INTO OPTIONS VALUES ('O040','Q010','D','E=mc^2');

INSERT INTO OPTIONS VALUES ('O041','Q011','A','8');
INSERT INTO OPTIONS VALUES ('O042','Q011','B','6');
INSERT INTO OPTIONS VALUES ('O043','Q011','C','12');
INSERT INTO OPTIONS VALUES ('O044','Q011','D','14');

INSERT INTO OPTIONS VALUES ('O045','Q012','A','H2O');
INSERT INTO OPTIONS VALUES ('O046','Q012','B','CO2');
INSERT INTO OPTIONS VALUES ('O047','Q012','C','NaCl');
INSERT INTO OPTIONS VALUES ('O048','Q012','D','O2');

INSERT INTO OPTIONS VALUES ('O049','Q013','A','6');
INSERT INTO OPTIONS VALUES ('O050','Q013','B','7');
INSERT INTO OPTIONS VALUES ('O051','Q013','C','8');
INSERT INTO OPTIONS VALUES ('O052','Q013','D','14');

INSERT INTO OPTIONS VALUES ('O053','Q014','A','Central Processing Unit');
INSERT INTO OPTIONS VALUES ('O054','Q014','B','Computer Personal Unit');
INSERT INTO OPTIONS VALUES ('O055','Q014','C','Core Processing Utility');
INSERT INTO OPTIONS VALUES ('O056','Q014','D','Central Program Utility');
```

```
INSERT INTO OPTIONS VALUES ('O057','Q015','A','Java');
INSERT INTO OPTIONS VALUES ('O058','Q015','B','C');
INSERT INTO OPTIONS VALUES ('O059','Q015','C','Python');
INSERT INTO OPTIONS VALUES ('O060','Q015','D','Assembly');

INSERT INTO OPTIONS VALUES ('O061','Q016','A','1001');
INSERT INTO OPTIONS VALUES ('O062','Q016','B','1010');
INSERT INTO OPTIONS VALUES ('O063','Q016','C','1100');
INSERT INTO OPTIONS VALUES ('O064','Q016','D','0110');

INSERT INTO OPTIONS VALUES ('O065','Q017','A','Open Oriented Processing');
INSERT INTO OPTIONS VALUES ('O066','Q017','B','Object Oriented Programming');
INSERT INTO OPTIONS VALUES ('O067','Q017','C','Optimized Object Protocol');
INSERT INTO OPTIONS VALUES ('O068','Q017','D','Output Oriented Program');

INSERT INTO OPTIONS VALUES ('O069','Q018','A','A duplicate key for joins');
INSERT INTO OPTIONS VALUES ('O070','Q018','B','Unique identifier for a record');
INSERT INTO OPTIONS VALUES ('O071','Q018','C','An index on a column');
INSERT INTO OPTIONS VALUES ('O072','Q018','D','A foreign reference');

INSERT INTO OPTIONS VALUES ('O073','Q019','A','Structured Query Language');
INSERT INTO OPTIONS VALUES ('O074','Q019','B','Simple Query Logic');
INSERT INTO OPTIONS VALUES ('O075','Q019','C','Sequential Query Language');
INSERT INTO OPTIONS VALUES ('O076','Q019','D','Standard Query Layout');

INSERT INTO OPTIONS VALUES ('O077','Q020','A','WHERE');
INSERT INTO OPTIONS VALUES ('O078','Q020','B','HAVING');
INSERT INTO OPTIONS VALUES ('O079','Q020','C','GROUP BY');
```

```
INSERT INTO OPTIONS VALUES ('O080','Q020','D','ORDER BY');

INSERT INTO OPTIONS VALUES ('O081','Q021','A','Unique identifier of its own table');
INSERT INTO OPTIONS VALUES ('O082','Q021','B','References primary key of another table');
INSERT INTO OPTIONS VALUES ('O083','Q021','C','An index used for sorting');
INSERT INTO OPTIONS VALUES ('O084','Q021','D','A composite key');

INSERT INTO OPTIONS VALUES ('O085','Q022','A','Organizing data to reduce redundancy');
INSERT INTO OPTIONS VALUES ('O086','Q022','B','Encrypting database tables');
INSERT INTO OPTIONS VALUES ('O087','Q022','C','Splitting a database into servers');
INSERT INTO OPTIONS VALUES ('O088','Q022','D','Adding indexes to tables');

INSERT INTO OPTIONS VALUES ('O089','Q023','A','Deleting matching rows');
INSERT INTO OPTIONS VALUES ('O090','Q023','B','Combining rows from two or more tables');
INSERT INTO OPTIONS VALUES ('O091','Q023','C','Filtering grouped results');
INSERT INTO OPTIONS VALUES ('O092','Q023','D','Sorting query output');

INSERT INTO OPTIONS VALUES ('O093','Q024','A','First Normal Form');
INSERT INTO OPTIONS VALUES ('O094','Q024','B','Second Normal Form');
INSERT INTO OPTIONS VALUES ('O095','Q024','C','Third Normal Form');
INSERT INTO OPTIONS VALUES ('O096','Q024','D','BCNF');

INSERT INTO OPTIONS VALUES ('O097','Q025','A','Atomicity Consistency Isolation Durability');
INSERT INTO OPTIONS VALUES ('O098','Q025','B','Access Control Integrity Data');
INSERT INTO OPTIONS VALUES ('O099','Q025','C','Availability Concurrency Integrity Durability');
INSERT INTO OPTIONS VALUES ('O100','Q025','D','Atomicity Concurrency Isolation Definition');

COMMIT;
```

```
UPDATE QUESTIONS SET correct_option_id = 'O001' WHERE question_id = 'Q001';
UPDATE QUESTIONS SET correct_option_id = 'O006' WHERE question_id = 'Q002';
UPDATE QUESTIONS SET correct_option_id = 'O010' WHERE question_id = 'Q003';
UPDATE QUESTIONS SET correct_option_id = 'O014' WHERE question_id = 'Q004';
UPDATE QUESTIONS SET correct_option_id = 'O018' WHERE question_id = 'Q005';
UPDATE QUESTIONS SET correct_option_id = 'O022' WHERE question_id = 'Q006';
UPDATE QUESTIONS SET correct_option_id = 'O026' WHERE question_id = 'Q007';
UPDATE QUESTIONS SET correct_option_id = 'O030' WHERE question_id = 'Q008';
UPDATE QUESTIONS SET correct_option_id = 'O034' WHERE question_id = 'Q009';
UPDATE QUESTIONS SET correct_option_id = 'O038' WHERE question_id = 'Q010';
UPDATE QUESTIONS SET correct_option_id = 'O042' WHERE question_id = 'Q011';
UPDATE QUESTIONS SET correct_option_id = 'O045' WHERE question_id = 'Q012';
UPDATE QUESTIONS SET correct_option_id = 'O050' WHERE question_id = 'Q013';
UPDATE QUESTIONS SET correct_option_id = 'O053' WHERE question_id = 'Q014';
UPDATE QUESTIONS SET correct_option_id = 'O058' WHERE question_id = 'Q015';
UPDATE QUESTIONS SET correct_option_id = 'O062' WHERE question_id = 'Q016';
UPDATE QUESTIONS SET correct_option_id = 'O066' WHERE question_id = 'Q017';
UPDATE QUESTIONS SET correct_option_id = 'O070' WHERE question_id = 'Q018';
UPDATE QUESTIONS SET correct_option_id = 'O073' WHERE question_id = 'Q019';
UPDATE QUESTIONS SET correct_option_id = 'O078' WHERE question_id = 'Q020';
UPDATE QUESTIONS SET correct_option_id = 'O082' WHERE question_id = 'Q021';
UPDATE QUESTIONS SET correct_option_id = 'O085' WHERE question_id = 'Q022';
UPDATE QUESTIONS SET correct_option_id = 'O090' WHERE question_id = 'Q023';
UPDATE QUESTIONS SET correct_option_id = 'O095' WHERE question_id = 'Q024';
UPDATE QUESTIONS SET correct_option_id = 'O097' WHERE question_id = 'Q025';

COMMIT;
```

```
INSERT INTO QUIZZES VALUES ('QZ001','S001','SUB001',TO_DATE('2025-01-10','YYYY-MM-DD'),4,5);
INSERT INTO QUIZZES VALUES ('QZ002','S002','SUB001',TO_DATE('2025-01-10','YYYY-MM-DD'),3,5);
INSERT INTO QUIZZES VALUES ('QZ003','S003','SUB002',TO_DATE('2025-01-11','YYYY-MM-DD'),5,5);
INSERT INTO QUIZZES VALUES ('QZ004','S004','SUB002',TO_DATE('2025-01-11','YYYY-MM-DD'),2,5);
INSERT INTO QUIZZES VALUES ('QZ005','S005','SUB003',TO_DATE('2025-01-12','YYYY-MM-DD'),3,5);
INSERT INTO QUIZZES VALUES ('QZ006','S006','SUB005',TO_DATE('2025-01-13','YYYY-MM-DD'),4,5);
INSERT INTO QUIZZES VALUES ('QZ007','S007','SUB008',TO_DATE('2025-01-14','YYYY-MM-DD'),5,5);
INSERT INTO QUIZZES VALUES ('QZ008','S008','SUB008',TO_DATE('2025-01-14','YYYY-MM-DD'),3,5);
INSERT INTO QUIZZES VALUES ('QZ009','S009','SUB001',TO_DATE('2025-01-15','YYYY-MM-DD'),2,5);
INSERT INTO QUIZZES VALUES ('QZ010','S010','SUB005',TO_DATE('2025-01-15','YYYY-MM-DD'),4,5);
INSERT INTO QUIZZES VALUES ('QZ011','S001','SUB008',TO_DATE('2025-02-01','YYYY-MM-DD'),5,5);
INSERT INTO QUIZZES VALUES ('QZ012','S002','SUB005',TO_DATE('2025-02-01','YYYY-MM-DD'),3,5);
INSERT INTO QUIZZES VALUES ('QZ013','S003','SUB001',TO_DATE('2025-02-05','YYYY-MM-DD'),4,5);
INSERT INTO QUIZZES VALUES ('QZ014','S011','SUB002',TO_DATE('2025-02-06','YYYY-MM-DD'),1,5);
INSERT INTO QUIZZES VALUES ('QZ015','S012','SUB003',TO_DATE('2025-02-07','YYYY-MM-DD'),4,5);
INSERT INTO QUIZZES VALUES ('QZ016','S013','SUB008',TO_DATE('2025-02-08','YYYY-MM-DD'),5,5);
INSERT INTO QUIZZES VALUES ('QZ017','S014','SUB005',TO_DATE('2025-02-09','YYYY-MM-DD'),2,5);
INSERT INTO QUIZZES VALUES ('QZ018','S015','SUB001',TO_DATE('2025-02-10','YYYY-MM-DD'),3,5);
INSERT INTO QUIZZES VALUES ('QZ019','S001','SUB002',TO_DATE('2025-03-01','YYYY-MM-DD'),4,5);
INSERT INTO QUIZZES VALUES ('QZ020','S002','SUB003',TO_DATE('2025-03-02','YYYY-MM-DD'),5,5);
COMMIT;

INSERT INTO QUIZ_ANSWERS VALUES ('A001','QZ001','Q001','O001','Y');
INSERT INTO QUIZ_ANSWERS VALUES ('A002','QZ001','Q002','O006','Y');
INSERT INTO QUIZ_ANSWERS VALUES ('A003','QZ001','Q003','O010','Y');
INSERT INTO QUIZ_ANSWERS VALUES ('A004','QZ001','Q004','O013','N');
INSERT INTO QUIZ_ANSWERS VALUES ('A005','QZ001','Q005','O018','Y');
```

```
INSERT INTO QUIZ_ANSWERS VALUES ('A006','QZ007','Q019','O073','Y');

INSERT INTO QUIZ_ANSWERS VALUES ('A007','QZ007','Q020','O078','Y');

INSERT INTO QUIZ_ANSWERS VALUES ('A008','QZ007','Q021','O082','Y');

INSERT INTO QUIZ_ANSWERS VALUES ('A009','QZ007','Q022','O085','Y');

INSERT INTO QUIZ_ANSWERS VALUES ('A010','QZ007','Q025','O097','Y');

COMMIT;



INSERT INTO RESULTS VALUES ('R001','S001','SUB001','QZ001',4,5,TO_DATE('2025-01-10','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R002','S002','SUB001','QZ002',3,5,TO_DATE('2025-01-10','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R003','S003','SUB002','QZ003',5,5,TO_DATE('2025-01-11','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R004','S004','SUB002','QZ004',2,5,TO_DATE('2025-01-11','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R005','S005','SUB003','QZ005',3,5,TO_DATE('2025-01-12','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R006','S006','SUB005','QZ006',4,5,TO_DATE('2025-01-13','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R007','S007','SUB008','QZ007',5,5,TO_DATE('2025-01-14','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R008','S008','SUB008','QZ008',3,5,TO_DATE('2025-01-14','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R009','S009','SUB001','QZ009',2,5,TO_DATE('2025-01-15','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R010','S010','SUB005','QZ010',4,5,TO_DATE('2025-01-15','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R011','S001','SUB008','QZ011',5,5,TO_DATE('2025-02-01','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R012','S002','SUB005','QZ012',3,5,TO_DATE('2025-02-01','YYYY-MM-DD'));
```

```
INSERT INTO RESULTS VALUES ('R013','S003','SUB001','QZ013',4,5,TO_DATE('2025-02-05','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R014','S011','SUB002','QZ014',1,5,TO_DATE('2025-02-06','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R015','S012','SUB003','QZ015',4,5,TO_DATE('2025-02-07','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R016','S013','SUB008','QZ016',5,5,TO_DATE('2025-02-08','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R017','S014','SUB005','QZ017',2,5,TO_DATE('2025-02-09','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R018','S015','SUB001','QZ018',3,5,TO_DATE('2025-02-10','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R019','S001','SUB002','QZ019',4,5,TO_DATE('2025-03-01','YYYY-MM-DD'));

INSERT INTO RESULTS VALUES ('R020','S002','SUB003','QZ020',5,5,TO_DATE('2025-03-02','YYYY-MM-DD'));

COMMIT;



INSERT INTO CGPA_RECORDS VALUES ('CG001','S001','MARKS_BASED',3.75,TO_DATE('2025-01-20','YYYY-MM-DD'));

INSERT INTO CGPA_RECORDS VALUES ('CG002','S002','SUBJECT_BASED',3.50,TO_DATE('2025-01-21','YYYY-MM-DD'));

INSERT INTO CGPA_RECORDS VALUES ('CG003','S003','MARKS_BASED',3.25,TO_DATE('2025-02-10','YYYY-MM-DD'));

INSERT INTO CGPA_RECORDS VALUES ('CG004','S007','SUBJECT_BASED',4.00,TO_DATE('2025-02-15','YYYY-MM-DD'));

INSERT INTO CGPA_RECORDS VALUES ('CG005','S013','MARKS_BASED',2.75,TO_DATE('2025-03-01','YYYY-MM-DD'));

COMMIT;
```

```
INSERT INTO CGPA_SUBJECTS VALUES ('CS001','CG001','SUB001',82,NULL,4.00);
INSERT INTO CGPA_SUBJECTS VALUES ('CS002','CG001','SUB002',76,NULL,3.75);
INSERT INTO CGPA_SUBJECTS VALUES ('CS003','CG001','SUB005',71,NULL,3.50);
INSERT INTO CGPA_SUBJECTS VALUES ('CS004','CG001','SUB004',65,NULL,3.25);

INSERT INTO CGPA_SUBJECTS VALUES ('CS005','CG002','SUB001',NULL,3.0,3.75);
INSERT INTO CGPA_SUBJECTS VALUES ('CS006','CG002','SUB002',NULL,3.0,3.50);
INSERT INTO CGPA_SUBJECTS VALUES ('CS007','CG002','SUB003',NULL,2.0,3.25);
INSERT INTO CGPA_SUBJECTS VALUES ('CS008','CG002','SUB004',NULL,2.0,3.00);

INSERT INTO CGPA_SUBJECTS VALUES ('CS009','CG003','SUB001',67,NULL,3.25);
INSERT INTO CGPA_SUBJECTS VALUES ('CS010','CG003','SUB003',60,NULL,3.00);
INSERT INTO CGPA_SUBJECTS VALUES ('CS011','CG003','SUB004',55,NULL,2.75);
INSERT INTO CGPA_SUBJECTS VALUES ('CS012','CG003','SUB006',50,NULL,2.50);

INSERT INTO CGPA_SUBJECTS VALUES ('CS013','CG004','SUB008',NULL,3.0,4.00);
INSERT INTO CGPA_SUBJECTS VALUES ('CS014','CG004','SUB005',NULL,3.0,4.00);
INSERT INTO CGPA_SUBJECTS VALUES ('CS015','CG004','SUB007',NULL,2.0,4.00);
INSERT INTO CGPA_SUBJECTS VALUES ('CS016','CG004','SUB001',NULL,3.0,4.00);

INSERT INTO CGPA_SUBJECTS VALUES ('CS017','CG005','SUB002',57,NULL,2.75);
INSERT INTO CGPA_SUBJECTS VALUES ('CS018','CG005','SUB003',50,NULL,2.50);
INSERT INTO CGPA_SUBJECTS VALUES ('CS019','CG005','SUB006',45,NULL,2.25);
INSERT INTO CGPA_SUBJECTS VALUES ('CS020','CG005','SUB004',42,NULL,2.00);

COMMIT;
```

## 🔥 1. Student Result Summary View

👉 Shows student performance per quiz

SQL

```sql
CREATE VIEW V_STUDENT_RESULTS AS
SELECT
    s.student_id,
    u.name,
    sub.subject_name,
    qz.quiz_id,
    qz.score,
    qz.total_marks,
    ROUND((qz.score/qz.total_marks)*100,2) AS percentage
FROM QUIZZES qz
JOIN STUDENTS s ON qz.student_id = s.student_id
JOIN USERS u ON s.user_id = u.user_id
JOIN SUBJECTS sub ON qz.subject_id = sub.subject_id;
```

✅ Used for:

- Result sheet
- Student dashboard

## 🔥 2. Question + Options + Correct Answer View

👉 Full question display (very useful)

SQL

```sql
CREATE VIEW V_QUESTION_DETAILS AS
SELECT
    q.question_id,
    q.question_text,
    o.option_label,
    o.option_text,
    CASE
        WHEN q.correct_option_id = o.option_id THEN 'CORRECT'
        ELSE 'WRONG'
    END AS status
FROM QUESTIONS q
JOIN OPTIONS o ON q.question_id = o.question_id;
```

✅ Used for:

- Showing quiz questions
- Review mode

## 🔥 3. Quiz Attempt Details View

👉 Shows what student selected vs correct

SQL

```sql
CREATE VIEW V_QUIZ_ANALYSIS AS
SELECT
    qa.quiz_id,
    qa.question_id,
    qa.selected_option_id,
    q.correct_option_id,
    qa.is_correct
FROM QUIZ_ANSWERS qa
JOIN QUESTIONS q ON qa.question_id = q.question_id;
```

✅ Used for:

- Answer review
- Performance analysis

## 🔥 4. Teacher Subject View

👉 Shows which teacher teaches which subject

SQL

```sql
CREATE VIEW V_TEACHER_SUBJECTS AS
SELECT
    t.teacher_id,
    u.name AS teacher_name,
    sub.subject_name
FROM SUBJECTS sub
JOIN TEACHERS t ON sub.teacher_id = t.teacher_id
JOIN USERS u ON t.user_id = u.user_id;
```

✅ Used for:

- Admin panel
- Teacher mapping

## 🔥 5. Top Performing Students View

👉 Real-world leaderboard

SQL

```sql
CREATE VIEW V_TOP_STUDENTS AS
SELECT
    s.student_id,
    u.name,
    ROUND(AVG(qz.score/qz.total_marks)*100,2) AS avg_percentage
FROM QUIZZES qz
JOIN STUDENTS s ON qz.student_id = s.student_id
JOIN USERS u ON s.user_id = u.user_id
GROUP BY s.student_id, u.name
ORDER BY avg_percentage DESC;
```

✅ Used for:

- Ranking
- Analytics

# TRIGGER

## 🔥 1. Limit Options Per Question

SQL

```sql
CREATE OR REPLACE TRIGGER LIMIT_OPTIONS_PER_QUESTION
BEFORE INSERT ON OPTIONS
FOR EACH ROW
DECLARE
    option_count NUMBER;
BEGIN
    SELECT COUNT(*) INTO option_count
    FROM OPTIONS
    WHERE question_id = :NEW.question_id;

    IF option_count >= 4 THEN
        RAISE_APPLICATION_ERROR(-20001, 'Only 4 options allowed per question');
    END IF;
END;
/
```

---

## 🔥 2. Require 4 Options Before Setting Correct Answer

SQL

```sql
CREATE OR REPLACE TRIGGER REQUIRE_4_OPTIONS_BEFORE_CORRECT
BEFORE UPDATE OF correct_option_id ON QUESTIONS
FOR EACH ROW
DECLARE
    option_count NUMBER;
BEGIN
    SELECT COUNT(*) INTO option_count
    FROM OPTIONS
    WHERE question_id = :NEW.question_id;

    IF option_count < 4 THEN
        RAISE_APPLICATION_ERROR(-20002, 'Add 4 options before setting correct answer');
    END IF;
END;
/
```

## 🔥 3. Automatically Mark Answer Correct or Wrong

```sql
CREATE OR REPLACE TRIGGER AUTO_SET_CORRECT_FLAG
BEFORE INSERT ON QUIZ_ANSWERS
FOR EACH ROW
DECLARE
    correct_opt VARCHAR2(10);
BEGIN
    SELECT correct_option_id INTO correct_opt
    FROM QUESTIONS
    WHERE question_id = :NEW.question_id;

    IF :NEW.selected_option_id = correct_opt THEN
        :NEW.is_correct := 'Y';
    ELSE
        :NEW.is_correct := 'N';
    END IF;
END;
/
```

---

## 🔥 4. Prevent Invalid Score Entry

```sql
CREATE OR REPLACE TRIGGER PREVENT_INVALID_SCORE
BEFORE INSERT OR UPDATE ON QUIZZES
FOR EACH ROW
BEGIN
    IF :NEW.score > :NEW.total_marks THEN
        RAISE_APPLICATION_ERROR(-20003, 'Score cannot be greater than total marks');
    END IF;
END;
/
```

## 🔥 5. Auto Fill Result Date

SQL

```sql
CREATE OR REPLACE TRIGGER AUTO_SET_RESULT_DATE
BEFORE INSERT ON RESULTS
FOR EACH ROW
BEGIN
    IF :NEW.result_date IS NULL THEN
        :NEW.result_date := SYSDATE;
    END IF;
END;
/
```

# PROCEDURE:

## 🔥 1. Add New Student

SQL

```sql
CREATE OR REPLACE PROCEDURE ADD_NEW_STUDENT (
    p_user_id VARCHAR2,
    p_name VARCHAR2,
    p_contact VARCHAR2,
    p_password VARCHAR2,
    p_student_id VARCHAR2
)
IS
BEGIN
    INSERT INTO USERS VALUES (p_user_id, p_name, p_contact, p_password, 'STUDENT');
    INSERT INTO STUDENTS VALUES (p_student_id, p_user_id);
END;
/
```

👉 Purpose: Add student + user together

---

## 🔥 2. Add New Teacher

SQL

```sql
CREATE OR REPLACE PROCEDURE ADD_NEW_TEACHER (
    p_user_id VARCHAR2,
    p_name VARCHAR2,
    p_contact VARCHAR2,
    p_password VARCHAR2,
    p_teacher_id VARCHAR2
)
IS
BEGIN
    INSERT INTO USERS VALUES (p_user_id, p_name, p_contact, p_password, 'TEACHER');
    INSERT INTO TEACHERS VALUES (p_teacher_id, p_user_id);
END;
/
```

👉 Purpose: Add teacher + user

## 🔥 3. Add Question

```sql
CREATE OR REPLACE PROCEDURE ADD_QUESTION (
    p_qid VARCHAR2,
    p_sid VARCHAR2,
    p_text VARCHAR2
)
IS
BEGIN
    INSERT INTO QUESTIONS (question_id, subject_id, question_text)
    VALUES (p_qid, p_sid, p_text);
END;
/
```

👉 Purpose: Insert question easily

---

## 🔥 4. Add Option

```sql
CREATE OR REPLACE PROCEDURE ADD_OPTION (
    p_oid VARCHAR2,
    p_qid VARCHAR2,
    p_label CHAR,
    p_text VARCHAR2
)
IS
BEGIN
    INSERT INTO OPTIONS VALUES (p_oid, p_qid, p_label, p_text);
END;
/
```

👉 Purpose: Add options (trigger will control max 4)

## 🔥 5. Set Correct Answer

SQL

```sql
CREATE OR REPLACE PROCEDURE SET_CORRECT_OPTION (
    p_qid VARCHAR2,
    p_oid VARCHAR2
)
IS
BEGIN
    UPDATE QUESTIONS
    SET correct_option_id = p_oid
    WHERE question_id = p_qid;
END;
/
```

👉 Purpose: Assign correct answer

## 🔥 6. Calculate and Update Quiz Score

SQL

```sql
CREATE OR REPLACE PROCEDURE CALCULATE_QUIZ_SCORE (
    p_quiz_id VARCHAR2
)
IS
    v_score NUMBER;
    v_total NUMBER;
BEGIN
    SELECT COUNT(*) INTO v_score
    FROM QUIZ_ANSWERS
    WHERE quiz_id = p_quiz_id AND is_correct = 'Y';

    SELECT COUNT(*) INTO v_total
    FROM QUIZ_ANSWERS
    WHERE quiz_id = p_quiz_id;

    UPDATE QUIZZES
    SET score = v_score,
        total_marks = v_total
    WHERE quiz_id = p_quiz_id;
END;
/
```

👉 Purpose: Auto calculate quiz result